

Seak Energetics' Luminode SDM-DIG-M module is designed for two-way communication of luminaires with a two-way DALI bus driver and control unit over the powerline. The module acts as a converter between the QM-50-SS13 and DALI protocols. It allows users to define lighting schedules along with different intensity levels, lag and fade time when motion is detected. The technology also allows intensity control according to DIN input settings and lighting regulation.

Smart city

Smart city projects are going strong all over the world. Any possible form of data from our actions in daily lives can be utilised to create optimised practical solutions in real-time scenarios.

One such solution is Hitachi's video-analytics system. Unlike general video systems that work based on pre-programmed logics with pre-determined solutions, it performs real-time analysis of live videos captured and thus delivers real-time solutions on the go. It is quite useful for queue management in airports, identity protection in medical facilities and classified areas, and clear imaging in unclear weather conditions.

Security

Offices and home premises need uncompromised security. Surebell home security system by Pictave is an app-based door answering and security system that lets users answer the door in real time from any location. It video-records any kind of motion within a preset radius, as well as audio and video of the person communicating through the answering machine. The recorded data is stored in the cloud, so users can access the data from anywhere. The system costs between ₹13,500 and 25,000.

For digital forensic professionals, Forensics Guru has come up with an innovative solution called



the WriteProtect Desktop Logicube that provides fast, secure, read-only write-blocking of suspect hard drives.

Smart transportation

For fuel saving and improved performance of vehicles on road, Hybridtronics has developed a complete hybrid kit that can retrofit in any bus and convert it into a hybrid bus, i.e., diesel/CNG + electric. To ensure cost saving on each kilometre covered, the Hybro 160 increases fuel economy by up to 30 per cent in heavy traffic. When fitted in light commercial vehicles such as tempo travellers, it reduces the cost per kilometre by 300 per cent.

Yet another interesting solution is the free bus Wi-Fi service by Maipu, which provides high-quality smart bus Internet service for travellers. The smart bus solution contains the core platform formed by multiple systems and the matching bus terminal, using 3G/4G lines of the carrier to provide multiple service modes such as bus video, Wi-Fi and local content.

Environment monitoring

It is essential for industries to maintain a healthy, safe and pollution-free environment inside and outside their premises. This can be achieved by continuous monitoring and analysis of the environmental data.

Phoenix Robotics has come up with an online environment monitoring tool called Aurassure, which helps to monitor, gather, analyse and forecast crucial information on

industry effluents and emissions. This solution greatly cuts down the need of on-site surveillance visits for industries and operational costs for maintaining analysts.

Various sensors and transmitters are

connected to the analyser to monitor the pollution parameters. The collected data is stored in the cloud in a secure encrypted format, which users can access through an interactive online web interface and mobile application.

Solar solutions

Paving the way for flexible solar panels, Enkay Technologies has developed a bendable glass-less solar panel that is based on 4-busbar technology. The company has completed its R&D and will soon launch the product in the domestic market for commercial purposes.

In India, flexible panels are a brand new concept and architects are planning to use these on facades to serve the dual purpose of shadowing and lighting. Apart from this, these panels can also be used in e-rickshaws and all-in-one street lighting.

Another highlight in the smart solar segment is the CST H3500 solar thermal solution by Qadsun. This unique heat exchanger provides instant heat and speeds up drying by making use of its four highly reflective mirrors. The mirrors are solar-grade and the entire structure is CED-coated.

The setup is ideal for temperature requirements greater than 160°C. It is a cost-effective solar heating solution for wastewater management, desalination plants, textile industry, leather industry, agriculture, etc. **EFY**

—Baishakhi Dutta, senior business journalist, EFY

If you want to share such case studies with EFY readers, please contact us at smartworld@efy.in